

2D 360° SURVEY LIDAR · LR-1F

360° AT FIFTY OLEI LR-1F SURVEY LIDAR

A full-circle scanner with 50 metres of reach and 0.06° angular resolution, built for hybrid navigation — reflectors where the site has them, natural features where it doesn't — on a motor designed for fast, bumpy duty.



■ WHY THIS SENSOR

- **360° full-angle scanning to 50 m** — no blind sector to design around and no second sensor to buy
- **0.06° angular resolution at 10 Hz** — 0.09° at 15 Hz, 0.12° at 20 Hz, 0.15° at 25 Hz — pick resolution or rate
- **60 000 points per second** — azimuth, distance and reflection strength on every return
- **Reflector and natural navigation** — one sensor covers a hybrid site, so the map doesn't dictate the hardware
- **< ±20 mm absolute accuracy on common targets** — < ±40 mm on reflectors, < 20 mm repeat
- **Motor built for speed and vibration** — specified for high-speed, bumpy operating environments
- **IP66, -10 to +50 °C, 900 g** — the highest ingress rating in the 2D line — wash-down and outdoor yards
- **M12 power and LAN, OLAMViewer and a ROS driver** — 5-pin A-coded supply, 8-pin A-coded LAN, 100 Mbps Ethernet

0.2–50 m

MEASUREMENT RANGE

360°

FIELD OF VIEW

0.06°

ANGLE RESOLUTION

60K

POINTS / SECOND

■ APPLICATION SCENARIOS



SURVEY & MAPPING

360° natural-feature and reflector mapping at range



HYBRID NAVIGATION

Reflector and natural navigation from one sensor



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



INTRALOGISTICS

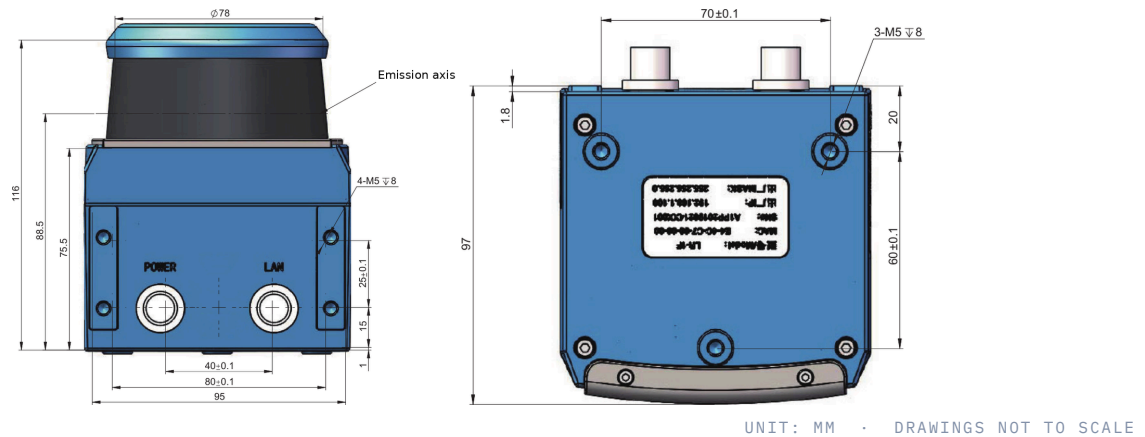
Forklifts, tuggers and conveyor zones in active aisles

LR-1F — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	360°
Beam divergence	5.8 mrad longitudinal 2.6 mrad transversal
Scanning frequency	10–25 Hz
Angle resolution (horizontal)	0.06° @ 10 Hz · 0.09° @ 15 Hz 0.12° @ 20 Hz · 0.15° @ 25 Hz
Measurement range (based on reflectivity)	0.2–50 m @ 80% reflectivity 0.2–15 m @ 10% · 0.2–5 m @ 4% · 0.2–50 m @ reflector
Accuracy	Common targets: absolute < ±20 mm Reflector: absolute < ±40 mm · Repeat < 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	60K pts/s
Horizontal error (horizontal angle)	≤ ±1°

PARAMETER	TECHNICAL DATA
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M12×1 A-coded (supply) 8-pin M12×1 A-coded socket (LAN)
Operating voltage	12–24 V DC
Power consumption @ 25 °C	7 W @ 15 Hz
Mechanical dimensions	95 × 97 × 116 mm
Weight	900 g
Degree of protection	IP66
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Optical indicators	RGB × 4
Ambient light limit	80 000 lux
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. Angular resolution and scan rate trade against each other — 0.06° is available at 10 Hz only. The 50 m figure is the 80% reflectivity and reflector case; plan on 15 m at 10% reflectivity and 5 m at 4%. Accuracy figures are standard values and depend on environmental conditions.

Installation. Bottom or back mounting. A 360° sensor needs its full circle clear — mount it above the platform's own structure rather than recessed into it, or the blind sectors will be created by the vehicle instead of the sensor. Keep the optical window clear of spray and dust and clean it on a schedule in dirty environments.

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